

Transformer Models for Question Answering on Autism Spectrum Disorder QA Dataset

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Abstract. Question answering (QA) Transformer-based models might become efficient in inclusive education. For example, one can test and tune such models with small closed-domain datasets before the implementation of a new system in an inclusive organization. However, studies in the sociomedical domain show that such models can be unpredictable. They can mislead a user or evoke aversive emotional states. The paper addresses the problem of investigating safety-first QA models that would generate user-friendly outputs. The study aims to analyze the performance of SOTA Transformer-based QA models on a custom dataset collected by the author of the paper. The dataset contains 1 134 question-answer pairs about autism spectrum disorders (ASD) in Russian. The study presents the validation and evaluation of extractive and generative QA models. The author used transfer learning techniques to investigate domain-specific QA properties and suggest solutions that might provide higher QA efficiency in the inclusion. The study shows that although generative QA models can misrepresent facts and generate false tokens, they might bring diversity in the system outputs and make the automated QA more user-friendly for younger people. Although extractive QA is more reliable, according to the metric scores presented in this study, such models might be less efficient than generative ones. The principal conclusion of the study is that a combination of generative and extractive approaches might lead to higher efficiency in building QA systems for inclusion. However, the performance of such combined systems in the inclusion is yet to be investigated.

Keywords: Question answering · Dialogue system · Transformer

1 Introduction

Closed-domain question answering (CDQA) systems might find an application in the sphere of inclusive education by providing information about inclusion

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and people with special needs in a user-friendly way. In general, CDQA is a challenging task due to the volume limitations of available datasets. In the past few years, much research has focused on the abilities of the Transformer-based [1] models that can achieve state-of-the-art (SOTA) results on various NLP tasks [2], including generative [3] and extractive question answering [4], even with limited training data.

Neural approaches allow us to achieve high metric scores. Nevertheless, the evaluation of the safety level of a built model remains unclear. The safety evaluation should help us to ensure the model does not misrepresent the data or generate false facts. Understanding how to evaluate and achieve the safety of a CDQA model for inclusive education is essential. A new app must not give misleading information, cause anxiety or negative emotions. One of the possible solutions might lie in the structure of a training dataset. For example, one can use or build a dataset containing different types of questions [5] or special keywords [6].

The purpose of the study was to explore the properties of state-of-the-art (SOTA) Transformer-based CDQA models fine-tuned for the autism spectrum disorder domain by comparing results of training extractive and generative QA systems on a dataset compiled by the author of the paper with results of other open- or medical (COVID-19 question answering) domain studies. The dataset presented in the study focuses on providing objective information about autism spectrum disorder and Asperger syndrome in the form of sets of reading passages and corresponding question-answer pairs [7]. The principal findings based on the comparison of Transformer models concern the analysis of extractive and generative QA performance. The author also looks forward to combining two approaches in the context of inclusive education. The novelty of the research is that the author tries to find out if the results of the SOTA models would be similar in the autism spectrum disorder, open- or medical domains.

The author of the paper consulted with psychiatrists from Russia. They have described to her during the interview the main features of apps for autistic people. According to experts, many people with special needs find it difficult to perceive new information by ear. The perception of visual information is more comfortable for many people with autism (but this is not true for everyone). The adapted information should be clear, unambiguous, unloaded from unnecessary details. The author of the paper considered these domain-specific features during the dataset compilation described in Sect. 3.

2 Related Work

Applying natural language technologies to teaching, diagnostics, rehabilitation of people with autism spectrum disorder is an extensively discussed topic. For example, in the 2000s, Boris Galitsky proposed a question answering system for autistic children [9]. His model aimed to teach patients reasoning about mental states. The online agent asked a user about mental states of characters from given scenarios. Some later studies proposed using systems based on closed-domain question answering to diagnose autism spectrum disorder. For

example, the Aquabot, developed in 2017, has a psychiatrist mode that asks a user questions to predict possible achluophobia or autism [10]. Another example is the Autism AI mobile app developed in 2020 that is based on a convolutional classifier and sets of questionnaires and can be used to predict possible autism spectrum disorder [11].

The research has focused on issues of building Conversational AI (ConvAI) models. ConvAI approaches are closely related to the development of natural language interfaces and human-machine interaction. Question answering, as one of the fundamental ConvAI tasks, copes with input questions within one or several domains by generating, extracting, or retrieving an answer based on the information from a data or knowledge base [8].

Closed-domain ConvAI systems have their roots in the 1960s. For instance, ELIZA [12], a closed-domain imitator of a psychotherapist session and the forerunner of modern chatbots, could generate dialogue lines by pattern matching and memory accessing. Another example is LUNAR [13], a closed-domain question answering system on moon geology and the forerunner of natural language interfaces based on syntax and semantic analyzers [14]. The parser of LUNAR was built with an augmented transition network (ATN), an automaton that changes states while moving from one word to another in a parsed piece of text.

Recent advances in natural language processing show that Transformer architecture based solely on attention mechanisms [1] allows achieving high metric scores on a wide range of tasks by transferring knowledge gained during the model training from one task to another through so-called fine-tuning [15]. For example, models based on Bidirectional Encoder Representations from Transformers (BERT) [27] fine-tuned for question answering, in fact, deal with Machine Reading Comprehension (MRC) task, which is to extract an answer from a given reading passage. In turn, QA models based on Generative Pre-trained Transformer [16] use zero-shot learning (a model learns to predict the data it has never seen before) to memorize the data and predict possible answers to questions.

According to the empirical findings from a study on applying GPT-2 [17] and GPT-3 [18] in building medical closed-domain question answering (CDQA) models, generative Transformer-based models might become harmfully biased [19] and, as a result, misrepresent information and cause negative emotions. On the contrary, extractive and retrieval-based approaches might be more reliable. Such models have already found their practical implementation in medical natural language processing tasks. Those are safety-first tasks [20].

The author of the paper has considered these hypotheses about generative and extractive question answering. To prove or disprove them, she has built and evaluated several CDQA models based on Transformer architecture using her custom dataset [7]. She aimed to find out domain-specific features of closed-domain question answering applied to the sphere of inclusive education. Although there are many domain-specific studies on medical natural language processing, the applicability of the existing approaches in inclusive education is not well investigated. That indicates a research gap which might be closed by further studies of the abilities and interpretability of Transformer-based models.

```
[
"title": "синдром Аспергера и РАС — общие сведения",
"paragraphs": [
    {
        "qas": [
            {
                "question": "Аутизм лечится?",
                "answers": [
                    {
                        "text": "Аутизм является пожизненным
                        "answer_start": 224,
                        "answer_end": 262
                    }
                ],
                "is_impossible": false
            },
        ],
        "context": "Этот текст подготовила группа аутичных в
    },
```


Fig. 1. A dataset sample.

3 Dataset

The Autism Spectrum Disorder Question Answering (ASD QA) [7] is a dataset compiled by the author of the paper for the present study. The dataset contains data from the online information resource for individuals with autism spectrum disorders (ASD) and Asperger syndrome [21], totaling 1 134 answered questions. One can use question-answer pairs and metadata from the dataset for training or fine-tuning models for machine reading comprehension (MRC), question answering (QA), text generation, etc.

The Stanford Question Answering Dataset (SQuAD) [22,23] was the source of inspiration for the ASD QA dataset [7]. Figure 1 presents a dataset sample. On the Fig. 1, *Title* denotes the name of a thematic cluster. The dataset includes three clusters. The first cluster comprises general information on autism spectrum disorder and Asperger syndrome. The second cluster contains facts about interaction and communication with people with special needs. The third cluster includes practical guidelines for parents. The block *Paragraphs* on the Fig. 1 denotes pairs of questions and answers (see *qas* on the Fig. 1) along with reading passages containing the corresponding information (see *context* on the Fig. 1). The dataset comprises 96 reading passages. The volume of the reading passages is 45 400 symbols, 6 578 words. The maximum length of a reading passage in the dataset is 512 symbols.

Each set of question-answer (QA) pairs (*qas*) includes one *question* and one *answer*. The volume of QA-sequences is 179 174 symbols, 26 269 words. Each set of *answers* includes their *texts* and metadata: (1) the numeric representation of an answer span in a corresponding *context* with its first symbol (*answer_start*)

Table 1. The dataset statistics.

	Train	Val	Test
Number of reading passages (E)	67	15	14
% of unanswerable questions (E)	5.8	3.95	4.35
Number of QA pairs	802	158	174
Number of tokens (word-level) in questions	5,524	1,204	1,381
Maximum length of a question (word-level)	20	15	15
Minimum length of a question (word-level)	2	2	2
Number of tokens (word-level) in answers	12,252	2,795	3,113
Maximum length of an answer (word-level)	45	50	42
Minimum length of an answer (word-level)	2	3	5
Maximum length of a reading passage (E)	512	512	512
Vocabulary size (E)	30,522	30,522	30,522
Out-of-vocabulary rate (E), mBERT	31.38	29.3	29.85
Out-of-vocabulary rate (E), mDBERT	0	0	0
Out-of-vocabulary rate (E), XLM-R	32.48	30.65	31.21
Out-of-vocabulary rate (E), ruBERT	0	0	0
Vocabulary size (G)	50,257	50,257	50,257
Out-of-vocabulary rate (G), GPT-2	0	0	0

and its last symbol (*answer_end*), and (2) a tag *is_impossible* that shows if a question can have a coherent answer. All the unanswerable questions in the ASD QA dataset are provided with the following plausible answer in Russian: *I cannot answer this question.*

The ASD QA dataset was shuffled and split into (1) a training set containing around 70% of total data, (2) a validation set that comprises around 15% of the data, and (3) a test set with around 15% of the data. The data was split with *train_test_split* method from Scikit-learn library [24]. Table 1 presents statistics on the split dataset. The table also shows specifications for extractive and generative question answering. *(E)* and *(G)* on the Table 1 denote respectively specific features of the dataset transformed for extractive and generative QA models. The table displays the out-of-vocabulary (OOV) rate separately for each Transformer-based model used in the study.

The ASD QA is a machine reading comprehension (MRC) dataset ready for building extractive QA models. The author has transformed the dataset for generative question answering: questions and answers were retrieved without metadata, and QA-pairs were supplemented with *start-* and *end-of-sentence* tags. The dataset was converted into a Python list object where each element represented a QA-pair. Figure 2 shows a dataset sample transformed for the training of generative QA models. $\langle s \rangle$ and $\langle /s \rangle$ denote start- and end-of-sentence tags respectively.

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<s> Расскажи мне сказку? Я не могу ответить на этот вопрос. </s>
<s> Что развивают у детей с аутизмом совместные игры с родителями? Если вы делаете
<s> Сможет ли мой ребенок с РАС заниматься спортом наравне с другими детьми? многие
<s> Что такое любовь? Я не могу ответить на этот вопрос. </s>
<s> Нужно ли бояться эпидемии аутизма? Нет никакой эпидемии аутизма. </s>
<s> Какими спортивными активностями можно заниматься вместе с ребенком с РАС? Забра
<s> Если я обижу аспи, она простит меня? Большинство Аспи могут легко прощать. </s>
```

Fig. 2. A dataset sample transformed for generative QA models training.

4 Models and Learning Configuration

Transformer architecture chosen for the experiments was described in [1]. This architecture is based solely on attention mechanisms with a fully connected feed-forward network (*FFN*). The position-wise FFN in the Transformer takes a vector x and passes it through the matrices *W1* and *W2* and bias vectors *b1* and *b2*:

$$FFN(x) = max(0, xW_1 + b_1)W_2 + b_2 \tag{1}$$

The unit of a multi-head self-attention mechanism A in the Transformer takes key-value pairs (K, V) as input. The keys are of dimension dk, and the values are of dimension dv. The attention function is multiplied by a matrix Q, and a *softmax* function is applied for the computation of the scaled dot-product attention described in [1]:

$$A(Q, K, V) = softmax(\frac{QK^T}{\sqrt{dk}})V \tag{2}$$

The Transformer-based models were fine-tuned using PyTorch [25] and HuggingFace [26] tools. Four models chosen for the experiments were pre-trained for masked language modeling (MLM) [27]. The author used these models for extractive QA. The list of models is as follows: Bidirectional Encoder Representations from Transformers multilingual (BERT, mBERT) base model (cased) [27], distilled version of BERT base multilingual (DBERT, mDBERT) model (cased) [28], Unsupervised Cross-lingual Representation Learning at Scale (XLM-RoBERTa, XLM-R) base model [29] and BERT base model for Russian (cased) fine-tuned by Geotrend (ruBERT) [30]. One of the models chosen for the experiments was pre-trained for traditional language modeling. The author used this model for generative QA. This model was Generative Pre-trained Transformer 2 [17].

The training data and model weights were stored on Google Cloud Storage [31]. The training was performed on Google Collaborative Environment [32] with Nvidia Tesla T4 graphics processing unit (GPU) [33]. During the experiments, the training time for each model was recorded in minutes. During the hyperparameters optimization, the author tuned batch size, learning rate, and the number of epochs. Each model was retrained ten times with different parameters.

The BERT-, DistilBERT- and XLM-RoBERTa-based models were retrained for 5, 10, 15, and 20 epochs with 1e−5, 3e−5, and 5e−5 learning rate. The optimal batch size for all those models was 1. The GPT-2-based model was retrained

for 15, 20, and 30 epochs with 1e−5, 3e−5, and 5e−5 learning rate. The optimal batch size for this model is 16. The optimal parameters are presented in the next section in Table 3 in the next section. Table 2 presents models' configurations.

Table 2. Models' configurations.

Parameter	BERT	DBERT	XLM-R	GPT-2
Dropout rate	0.1	0.1	0.1	0.1
Activation function	GELU	GELU	GELU	GELU
Hidden layers	12	6	12	12
Embeddings	512	512	512	768
Attention heads	12	12	12	12
Vocabulary size	30,522	30,522	30,522	50,257
Model parameters	110M	66M	125M	117M

5 Experiments and Results

One of the issues identified during the training is the "weight" of the latest natural language processing models. Even a small version of GPT-2 with 117 million parameters and BERT-based models with 110 million parameters necessitate high computation power to run training sessions. The issue was solved by using smaller batch sizes and applying a Python garbage collector. The garbage collector for the automatic memory allocation management prevented the crash of sessions caused by using all Random-Access Memory.

The next issue was related to the metric scores. The models mostly showed high precision and low recall, which means that the models were outputting very few but frequently correct answers. The problem was solved by increasing the dataset volume. During the experiments, it was found that shorter reading passages lead to better results. The reduction of reading passages' maximum length from 1 024 to 512 symbols increased the average model performance (F1-Score and Exact Match) by around 10%.

Table 3. Results obtained on ASD QA (extractive and generative question answering).

Model name	Time (minutes)	Number of epochs	Learning rate	Batch size	EM	F1
mBERT	22	10	3e−5	1	0.29	0.40
mDBERT	11	20	1e−5	1	0.32	0.42
XLM-R	10	10	3e−5	1	**0.39**	**0.48**
ruBERT	10	10	5e−5	1	0.30	0.39
GPT-2	30	30	3e−5	16	**0.41**	**0.53**

The final issue is the choice of models. The investigation of the HuggingFace Transformers repository showed that models pre-trained on masked language

modeling (MLM) and traditional language modeling (LM) are more flexible than narrow models pre-trained for question answering or machine reading comprehension (MRC). Narrow models could allow achieving higher metric scores after the fine-tuning. However, such models do not have zero-shot learning capabilities and have more dataset requirements. During the experiments, it was found that with low-resource closed-domain datasets models pre-trained on a target language data do not achieve higher results than multilingual ones. Table 3 shows the models' results.

6 Discussion

The author has compared the achieved metric scores (see Sect. 5) with the results of some similar studies. Table 4 is a comparison table of metric scores of Transformer-based models trained on open- and medical domain datasets. The properties of the dataset used in this study should differ from the features of the medical domain because texts about autism spectrum disorder (ASD) from the dataset are not professional. Such texts are usually being adapted for parents of people with ASD, neurotypical adults, adults with special needs, etc. In turn, medical texts that address professionals include more special vocabulary. Authors of such documents usually do not adapt their texts for non-professionals. Nevertheless, these domains have a lot in common because they both cover various medical topics. That is why the author decided to compare her results with the results of similar models trained on the medical COVID-19 Open Research Dataset (CORD-19) [34].

Table 4. Results obtained on open-domain and medical COVID-19 benchmarks.

Model name	Dataset	EM	F1
BERT	CORD-19 (medical, COVID-19)	81.5	88.3
DBERT	CORD-19 (medical, COVID-19)	80.6	87.3
BERT	SQuAD 1.1 dev (open, English)	81.3	88.7
DBERT	SQuAD 1.1 dev (open, English)	80.1	87.5
XLM-R	MKQA (open, multilingual)	–	46.0
mBERT	MKQA (open, multilingual)	–	44.1
mT5 (generative QA)	MKQA (open, multilingual)	–	38.5

Table 4 includes metric scores given in a paper describing COBERT, a BERT-based COVID-19 question answering system [35]. The authors of the COBERT compared the results of BERT-[27] and DistilBERT-based [28] models obtained on CORD-19 and open-domain SQuAD 1.1 [22] datasets in English. The results of multilingual BERT- and DistilBERT-based models obtained on the ASD QA dataset presented in this study differ. BERT showed slightly better results than

DistilBERT for both CORD-19 and SQuAD. In this study, a multilingual version of DistilBERT became more efficient than multilingual BERT on the ASD QA dataset (see Table 3).

The table of comparison includes F1 scores achieved by multi- and crosslingual Transformer-based models trained on Multilingual Open Domain Question Answering dataset (MKQA) [36]. The results presented in the MKQA paper allow us to compare the efficiency of generative and extractive question answering. The MKQA authors have chosen a multilingual version of T5 [37] Transformer-based model to implement the generative approach. They have also compared the efficiency of XLM-RoBERTa [29] and multilingual BERT. XLM-RoBERTa became the most powerful model. That is consistent with the results obtained on the ASD QA dataset. In this study, XLM-RoBERTa showed the highest metric scores among all the models chosen for the extractive approach implementation. In the MKQA case, the generative model achieved lower metric scores, whereas the same approach allowed the author to get the best results in this study.

7 Conclusion

Overall, in the study, the author compares SOTA Transformer-based question answering models' performance in the autism spectrum disorder domain with their performance in the open- or medical domain. The hypothesis that extractive models would cope with the task significantly better due to their reliability did not confirm according to the metric scores of the models and the analysis of the models' outputs obtained on the test data. The traditional language model fine-tuned for generative question answering showed higher results than any other model for extractive question answering used in the study.

Among the extractive QA models, the most effective became XLM-RoBERTa. Supposedly, the number of model parameters is the reason for that. The number of parameters is the highest for XLM-RoBERTa. It is also high for GPT-2. Apart from that, the number of embeddings is the highest for GPT-2. Out-of-vocabulary (OOV) rate linked to the vocabulary size did not play a crucial role (for example, XLM-RoBERTa had the highest OOV rate, but this model showed the highest performance among all the extractive models from the study). The author hasn't noticed any significant advantages of ruBERT model pre-trained on Russian (target language) data over multilingual ones.

A large number of unsuccessful outputs produced by trained systems shows that generative and extractive models are both yet far from the first trials of their integration into the education processes. One of the possible reasons for some unsuccessful outputs is the vocabulary of texts about autism spectrum disorders. Texts from the dataset used in the study contain specific words from the medical domain. For example, such texts comprise designations of mental, neurodevelopmental, generic, and other disorders (Asperger's syndrome, intellectual disability, Down's syndrome). They also include other medical terms (diagnosis, hypersensitivity, etc.). Those terms are mixed in the dataset with everyday

vocabulary about childhood and adolescence, for example, names of games and school-related words. Due to the topics' narrowness, the sentences in the dataset have a similar structure and are not lexically diverse.

Extractive models are considered to be more reliable than generative ones. They do not generate a new answer but extract one from a given piece of text. However, the study showed that such models could give misleading answers as often as generative models while being trained on a low-resource closed-domain dataset. The reason for that is that such models can shift the answer by extracting the right sentence from the reading passage but the wrong word or phrase due to the lexical similarities and little diversity in the dataset vocabulary. This problem can be solved by providing the dataset with shorter answers consisting of one or two words so that models would be able to learn to extract specific unique text pieces.

Generative models could generate correct information based on the knowledge from their memory not answering a user's question. The issue was caused by the large number of similar terms repeated in different reading passages and question-answer pairs. The models were so to say confused by a large amount of repeating vocabulary. Possibly, the problem can be solved by enriching the dataset with texts on different topics with a more diverse vocabulary.

In the longer term, the author decided to continue experimenting with the dataset transformations. In particular, the author plans to add more answers to each question and investigate the effect of the length of answers and questions on system performance. The author also intends to find out how various Transformer-based models deal with unanswerable questions and which techniques might lead to higher efficiency in solving this task.

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